# ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

**C01-3**

**EXPERIMENT 3**

**AIM:**

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

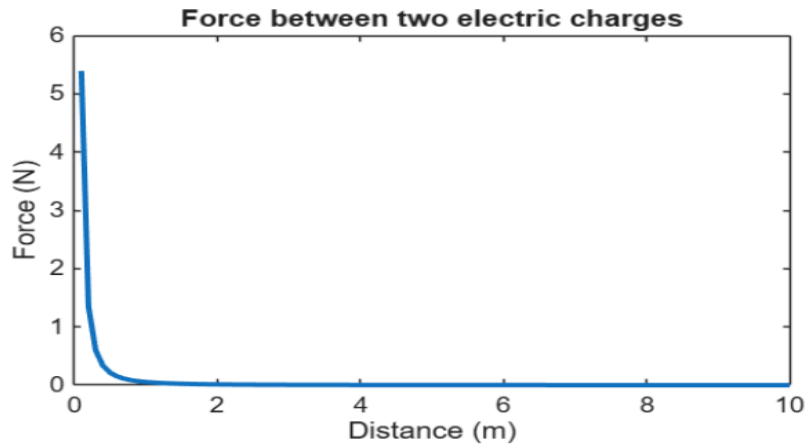
**Test case (i) Determination and plotting of force between two electric charges with varying distance**

**PROGRAM:**

```
clc;
clear;
q1 = input('Enter the charge q1=');
q2 = input('Enter the charge q2=');
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
r = linspace(0.1,10,100);
F = k*q1*q2./(r.^2);
disp('Force at r = 1 m is:');
disp(k*q1*q2/(1^2));
plot(r,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force (N)');
title('Force between two electric charges');
```

**OUTPUT:**

```
Force at r = 1 m is:
    0.0539
```

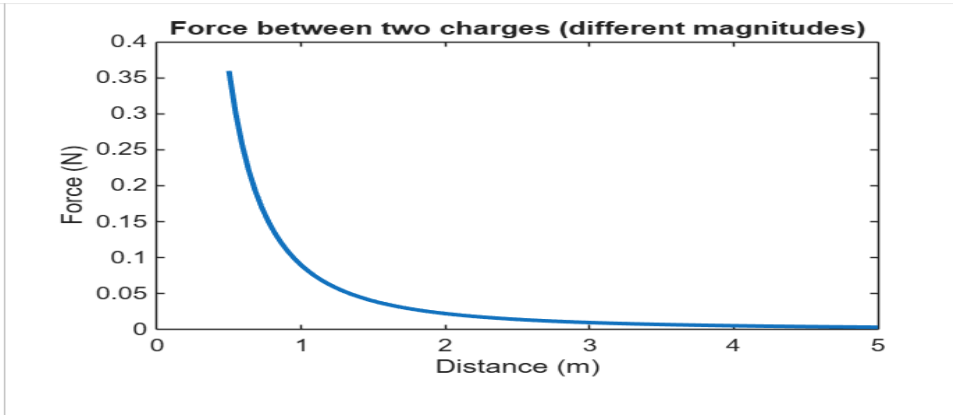


**Test case (ii)**
**Determination and plotting of force between two electric charges with different magnitudes**

**Program:**

```
clc;
q1 = 5e-6;
q2 = 2e-6;
epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);
r = linspace(0.5,5,100);
F = k*q1*q2./(r.^2);
plot(r,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force (N)');
title('Force between two charges (different magnitudes)');
```

**Output:**



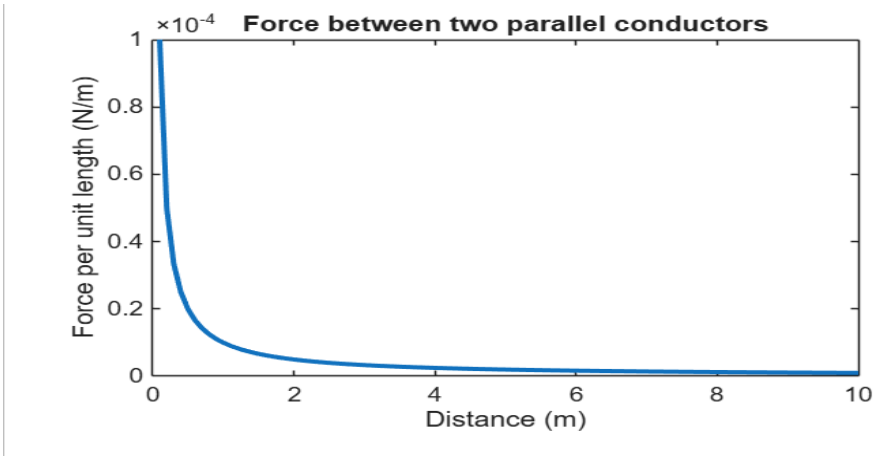

**Test case (iii) Determination and plotting of force per unit length between two parallel conductors with varying distance**

**Program:**

```
mu0 = 4*pi*1e-7;
I1 = 5;
I2 = 10;
d = linspace(0.1,10,100);
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
```

```
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
title('Force between two parallel conductors');
```

**OUTPUT:**

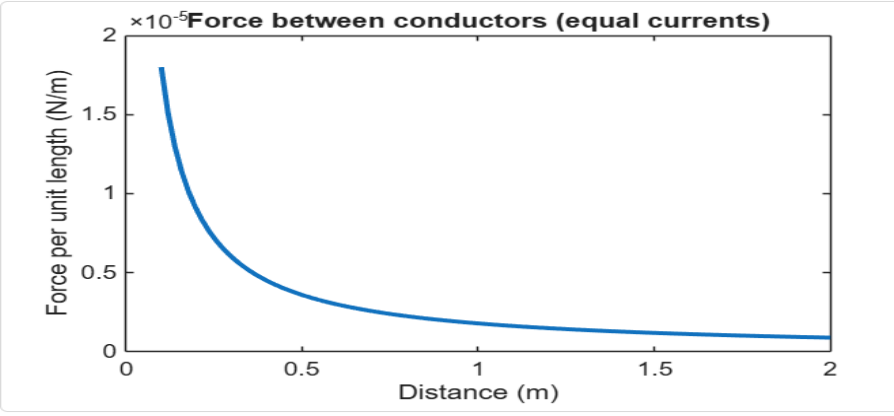



**Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents**

**Program:**

```
clc;
mu0 = 4*pi*1e-7;
I1 = 3;
I2 = 3;
d = linspace(0.1,2,100);
F = (mu0*I1*I2)./(2*pi*d);
plot(d,F,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Force per unit length (N/m)');
title('Force between conductors (equal currents)');
```

**OUTPUT:**




**RESULT:**

Thus the plot and determination of the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB

| Objective | Assessment Criterion | Excellent – 4 | Good – 3 | Satisfactory – 2 | Needs Improvement – 1 | Weightage | Marks given | Marks Obtained |
|---|---|---|---|---|---|---|---|---|
| Identify physical quantities and force models | Identify physical quantities and force models | Correctly identifies charge/current, separation, medium and force direction for both cases. | Identifies most quantities and models correctly. | Needs guidance in selecting the correct relation or direction. | Uses incorrect force models or quantities. | 25 | 2.5 | |
| Calculate electric and magnetic forces | Calculate electric and magnetic forces | Implements Coulomb-force and parallel-current force calculations correctly with units. | Calculations are mostly correct with minor errors. | Some calculations are correct but require assistance. | Force calculations are invalid or absent. | 25 | 2.5 | |
| Evaluate force versus separation | Evaluate force versus separation | Correctly varies separation, plots trends and checks theoretical distance dependence. | Trend and plots are correct with minor omissions. | Limited sweep or incomplete trend interpretation. | Does not establish a valid relationship. | 30 | 3 | |
| Explain and report results | Explain and report results | Explains attraction/repulsion and reports assumptions, units and conclusions clearly. | Conclusion is correct with minor documentation gaps. | Conclusion is incomplete or partly inaccurate. | Conclusion is missing or incorrect. | 20 | 2 | |
| **TOTAL** | | | | | | 100 | 10 | |